

NE 511: Multi-Physics of Nuclear Reactors

UNIT 3: Low Fidelity Multi-Physics Modeling

*Lecture 17: SPACE-ENERGY DEPENDENT DYNAMICS: QUASI-STATIC METHODS; KINETICS VS. DYNAMICS
DYNAMIC REACTIVITY COEFFICIENTS*

FLUX FACTORIZATION APPROACHES

	Model		
	Direct Numerical	Improved Quasi-static	Quasi-static
$\partial \psi / \partial t$	Yes	Yes	No
α / ν	Yes	Yes	Yes
$S_d[p(t')\psi(r, E, t')]$	Yes	Yes	Yes
Implicit coupling via feedback	Yes	Yes	Yes
Decoupled shape distortion	Yes	Yes	Yes
Initial shape	Yes	Yes	Yes
	Model		
	Adiabatic	Adiabatic with pre-calculated shape function	PRM
$\partial \psi / \partial t$	No	No	No
α / ν	No	No	No
$S_d[p(t')\psi(r, E, t')]$	No	No	No
Implicit coupling via feedback	Yes	No	No
Decoupled shape distortion	Yes	Yes	No
Initial shape	Yes	Yes	Yes

FLUX FACTORIZATION APPROACHES

- The direct numerical and improved quasi-static methods are mathematically equivalent in the sense that no approximations except finite differencing are introduced.
 - Thus, both methods yield the same result for sufficiently small time steps.
- The accuracy of the other methods generally decreases with decreasing sophistication of the shape function balance equation.
- Both adiabatic methods calculate the flux shape from the same type of balance equation, $(\mathbf{M} - \lambda \mathbf{F})\Psi = 0$.
 - The simpler of the two methods allows only changes in the reactor for which the sequence of configurations is known in advance.
 - For such known sequences of motion, the flux distortion and its effect on the reactivity can be pre-calculated.
 - The resulting time-dependent reactivity is inserted into the point kinetics equation.
 - If there are no other changes in the reactor except the pre-calculated configurations, the results may be fairly accurate.

INTERPRETATION OF THE QUASI-STATIC ASSUMPTIONS

The evaluation of the type of inaccuracy caused by the quasi-static calculation of the flux shape is based on a physical understanding of the approach.

Let's consider a transient following an asymmetrically applied step change in the reactor:

Example: Instantaneous withdrawal of an off-center control rod

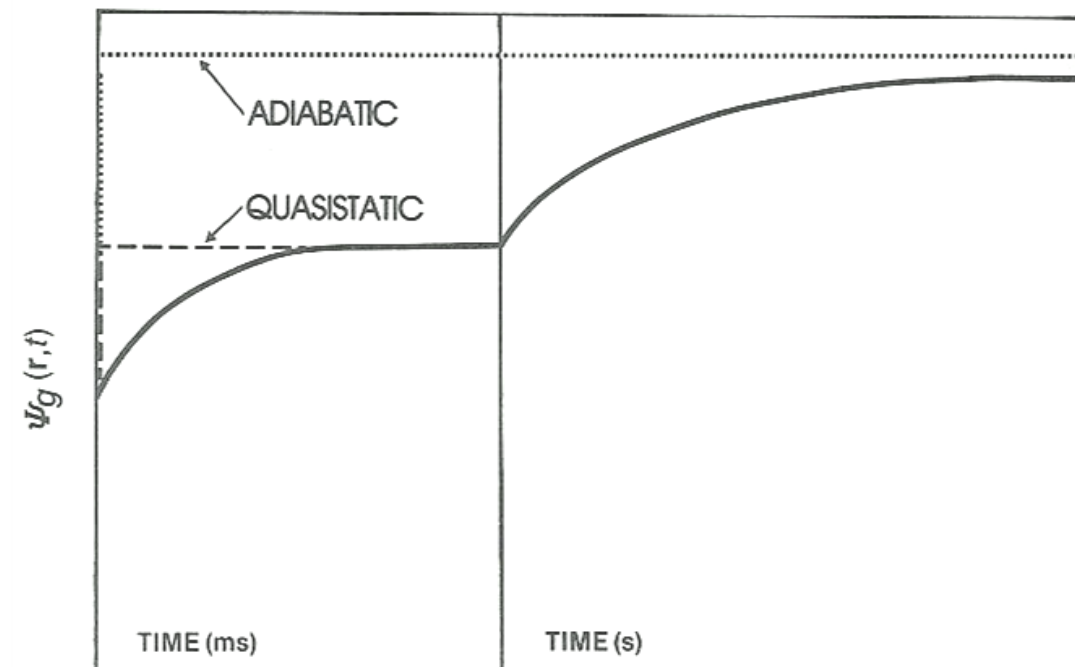
- shape function increases in the area where the neutron absorption is reduced;
- shape function decreases in the remaining area;
- however, the integral constraint condition must remain constant:

$$\int_V \int_0^\infty \frac{\phi_0^*(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dV dE = K_0$$

INTERPRETATION OF THE QUASI-STATIC ASSUMPTIONS

The increase of the shape function exhibits two distinct phases:

1. Very rapid (prompt) adjustment of the shape of the neutron population to the changed configuration.
2. Comparatively slow (delayed) shape change due to the adjustment of the precursor population.



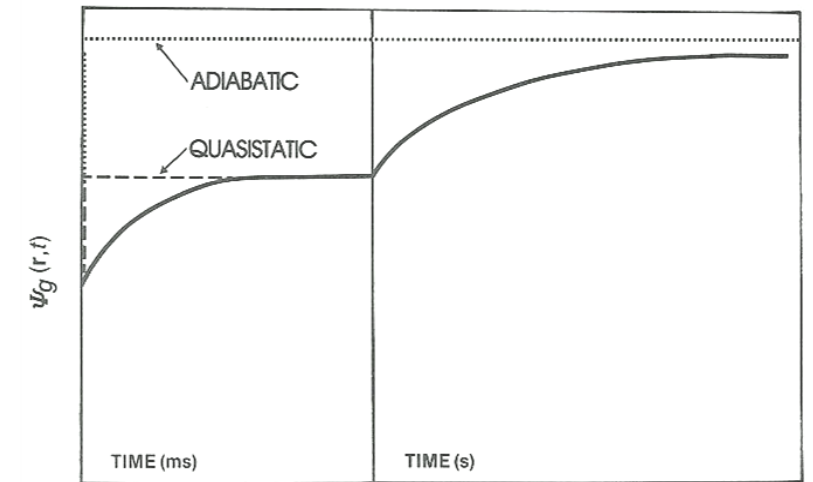
INTERPRETATION OF THE QUASI-STATIC ASSUMPTIONS

The prompt and the delayed response of the flux to a step change in the system correspond to the prompt and delayed changes of the amplitude.

The only difference is that:

- asymptotically the amplitude function varies exponentially;
- asymptotically the shape function becomes constant.

The temporary plateau of the flux shape is described as accurately by the PJA as the corresponding plateau of the amplitude transient.



The reason is - the ψ derivative term in the balance equation for the flux shape becomes practically negligible after the prompt shape jump due to the division by the large neutron velocity.

$$\left(F_p - M - \frac{1}{v \Delta t_s^s} - \frac{1}{v} \frac{\dot{p}}{p} \right) \psi(r, E, t^s) = -\frac{1}{p} S_d [p(t') \psi(r, E, t')] + \frac{\psi(r, E, t_{s-1}^s)}{v \Delta t_{s-1}^s}$$

After the prompt-jump, the shape function can be calculated from the equation with the ψ derivative deleted (from the balance equation of the quasi-static approach).

$$\left(F_p - M - \frac{1}{v} \frac{\dot{p}}{p} \right) \psi(r, E, t^s) = -\frac{1}{p} S_d [p(t') \psi(r, E, t')]$$

INTERPRETATION OF THE QUASI-STATIC ASSUMPTIONS

Further variation of the shape function is caused by the redistribution of the precursors initiated by the prompt variation of the flux shape.

The resulting time dependence of delayed neutron source is fully incorporated in the quasi-static approach.

Since the shape derivative term can be neglected during the delayed transition, the quasi-static approach describes the delayed shape transition properly.

Asymptotically, the shape derivative is exactly zero.

Thus, the quasi-static approach describes the asymptotic flux shape exactly.

INTERPRETATION OF THE ADIABATIC ASSUMPTION

During the prompt-jump following a step change in the reactor, the typical inaccuracy of the adiabatic approach is of the same type but larger than that of the quasi-static approach - the delayed transition is missing altogether in the adiabatic approach.

Integrated terms (inhour equation) are considered instead of the space-dependent terms in the correct balance equation, i.e.:

$$\sum_k \frac{\lambda_k}{\alpha + \lambda_k} \beta_k - \alpha \Lambda = \beta - \rho$$

If $\rho = 0$ and $\alpha = 0$, both sides of the above equation are reduced to β .

If $\rho \neq 0$, the question is whether the corresponding variation of the left side of the equation is achieved primarily by variation of the delayed neutron term or of the $\alpha \Lambda$ term, which corresponds to the α / ν term in the correct balance equation.

For $0 < \rho < \beta$, the $\alpha \Lambda$ term is very small:

Thus, the reduction of the left side of the equation is primarily due to the reduction of the delayed neutron source.

INTERPRETATION OF THE ADIABATIC ASSUMPTION

$$\sum_k \frac{\lambda_k}{\alpha + \lambda_k} \beta_k - \alpha \Lambda = \beta - \rho$$

Even for $\rho = \beta$, the $\alpha \Lambda$ term itself is still small:

Thus, its variation is small compared to the reduction of the delayed neutron term.

At $\rho = \beta$:

The delayed neutron source term in the equation is practically reduced to zero with virtually nothing compensated by the α / v absorption (represented by $\alpha \Lambda$ in this equation).

At $\rho = 2\beta$:

The variations of both terms are comparable.

The delayed neutron term has been reduced by about β ; i.e., from its original value of β to practically zero.

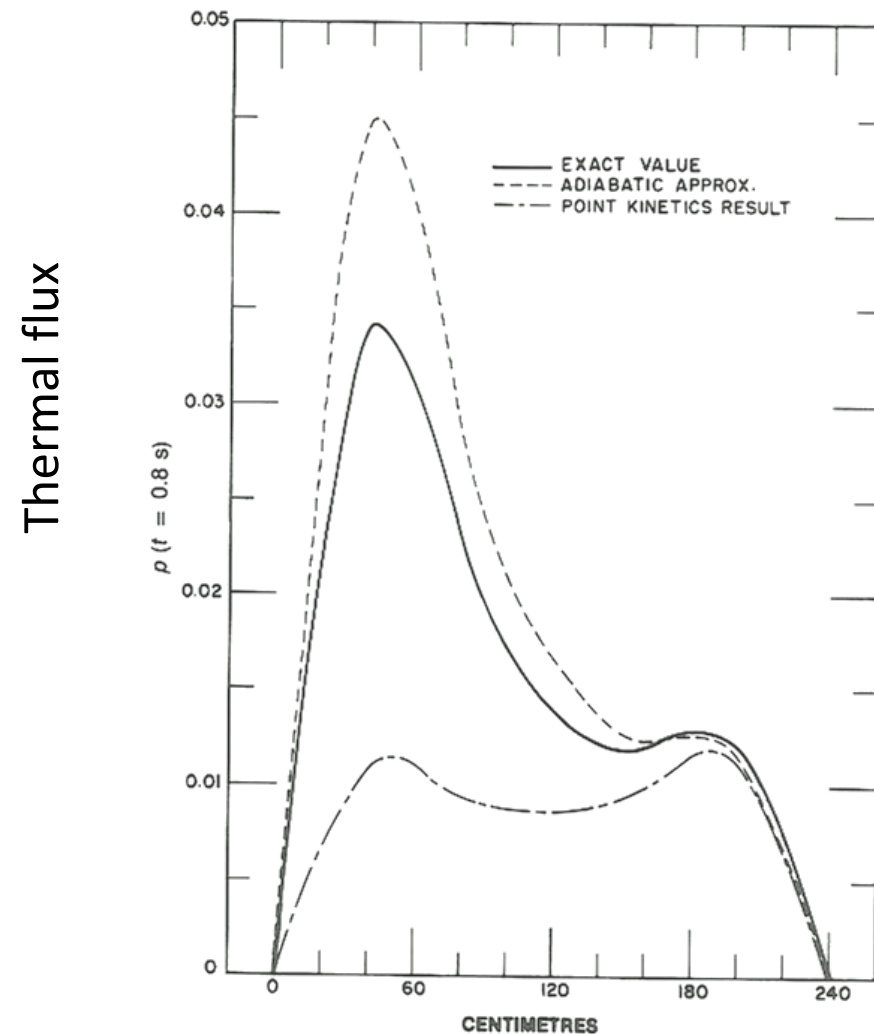
The $\alpha \Lambda$ term has also been changed by β , from its original value of zero to about β .

For $\rho = 2\beta$, the $\alpha \Lambda$ variation dominates.

Since for most transients ρ is much smaller than 2β , the reduction of the delayed neutron term in the λ mode equation dominates the α / v term in practically all transients.

INTERPRETATION OF THE ADIABATIC ASSUMPTION

In this example, an amount of fissile material ($\delta\rho < \beta$) is inserted in the left quarter of a thermal slab reactor model.



The instantaneous re-distribution of the precursors in the adiabatic approach strongly exaggerates the flux shape distortion.

In conclusion, the adiabatic approach is not accurate enough for thermal reactor dynamics.

The deviation is due to the neglect of the time delay in the adjustment of the precursor shape → the neglect of the shape derivative should have virtually no effect on the result.

QUASI-STATIC VS. ADIABATIC ASSUMPTION

It was proved numerically that the quasi-static approach, in which the precursor distribution is described correctly, yields an accurate treatment of spatial kinetics problems.

The importance of the precursor distribution for the flux shape can be understood by applying the fission chain concept → although the number of delayed neutrons is much smaller than the number of prompt neutrons, the delayed neutrons are the only source of fission chains.

Thus, the space distribution of the precursors determines the space distribution of the initiation of fission chains and, through it, affects the entire flux distribution.

The neutron velocities in fast reactors are larger than in thermal reactors → the neglect of the shape derivative should lead to smaller errors in fast than in thermal reactors.

This difference is not important in the improved quasi-static approach where the shape function derivative is accounted for.

Kinetics versus Dynamics

COMPARISON OF KINETICS AND DYNAMICS RESULTS

In realistic modeling of core-disruptive accidents:

1. The reactivity insertion rate obtained from an SED (space-energy dependent) treatment would generally be larger than in the PRM (point reactor model) due to an increased prediction of the initiating reactivity.
2. This in turn may lead to larger deviations of SED-type and PRM-type predictions for the overall energy release and even more so for energy increments.

COMPARISON OF KINETICS AND DYNAMICS RESULTS

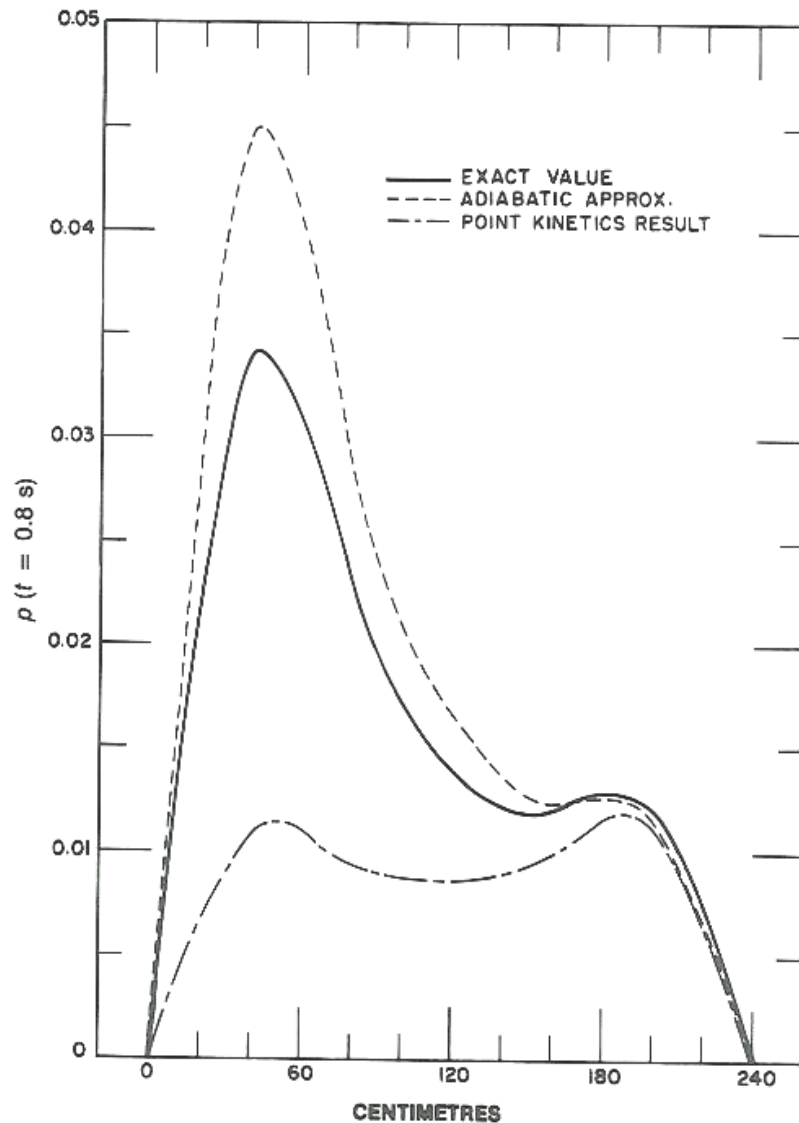
In the next comparisons:

- Transients for subprompt- and superprompt-critical reactivity insertions without and with feedback are considered.
- Results are calculated with different methods for thermal and fast reactors.
- Information on the effect of different neutron lifetimes on the accuracy of the quasi-static approach is provided.

First, let us recall that:

- The replacement of the prompt shape adjustment by an instantaneous one (quasi-static approach) may not be as good an approximation in thermal reactors as in fast reactors because of the lower thermal reactor average neutron velocity.
- This is, however, less important in the improved quasi-static approach.

Subprompt-Critical Thermal Reactor Kinetics Problem

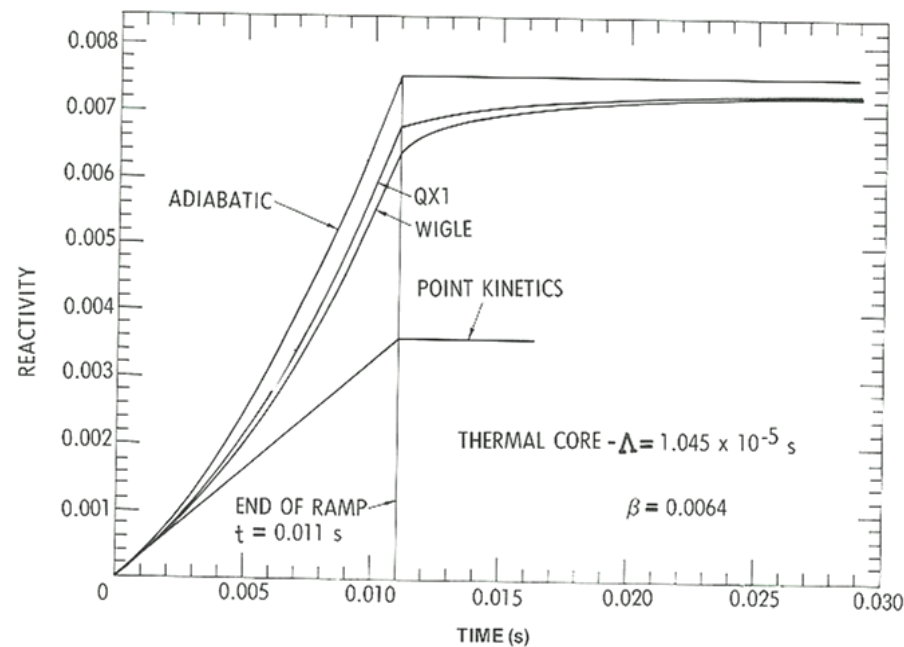


Delayed supercritical excursion thermal flux at 0.8 s for a 240-cm core

- The results of the [point reactor model](#) exhibit a large error due to the neglect of the considerable shape distortion.
- The [adiabatic results](#) show a sizable error due to the neglect of the delay in the adjustment of the precursor population.
- For $\rho < \beta$ the delay of the precursor population has a strong effect on the shape function through source multiplication.
- [Quasi-static results](#) should be practically indistinguishable from the finite difference results since the shape variation is very slow in this case.
- The results of the [improved quasi-static method](#) should be identical with the finite difference results.

*In similar **subprompt-critical transients in fast reactors**, all results should be closer together because of smaller shape distortions.*

Superprompt-Critical Ramp-Induced Transient in a Thermal Reactor



Reactivity versus time for limited ramp insertion in a thermal reactor core

- The reactivity is inserted such that it appears as a linear ramp in the point reactor model.
- The upper curve is obtained from the adiabatic treatment with an instantaneous adjustment of the complete flux shape.
- The next curve results from the quasi-static model in which only the prompt adjustment occurs instantaneously.
- Both the adiabatic and the quasi-static reactivity are higher than the correct reactivity on the rising part of the reactivity curve since they "anticipate" the flux deformation.

This situation would be reversed for decreasing reactivity, where the quasi-static and the adiabatic reactivity are below the correct reactivity.

*In **superprompt-critical fast reactor** transients, the adiabatic, the quasi-static, and the correct (WIGLE) results are close together because the flux shape is nearly completely determined by the prompt neutrons, since the delayed neutron source has lost its dominating effect.*

*For a **subprompt-critical reactivity insertion**, a set of curves would be obtained in a similar way for thermal as well as for fast reactors.*

Superprompt-Critical Ramp-Induced Transient in a Thermal Reactor

Comparison of Factorization Approaches
Amplitude Function at t = 0.022s for Ramp-Reactivity Insertion

Reactor Generation Time		Model				
		Direct Numerical	Improved Quasi-static	Quasi-static	Adiabatic	Point Reactor Model
Thermal core ($\Lambda = 1 \times 10^{-5} s$)	$p(0.022 s)$	14.92	14.92	16.18	21.27	2.253
	p / p_{WIGLE}	1.000	1.000	1.084	1.426	0.1510
					+ 8.4%	+ 43%
						- 85%

The flux amplitude results are in exact agreement between the values of the improved quasi-static and the direct numerical (finite difference and WIGLE) methods.

Superprompt-Critical Asymmetric Transient

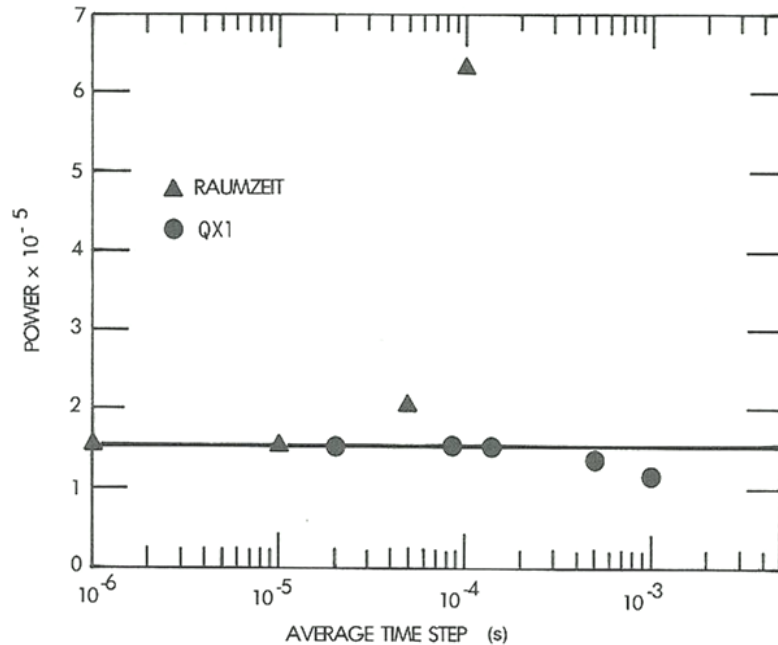
A set of one-dimensional kinetics benchmark problems has been solved with QX1, WIGLE, and RAUMZEIT.

WIGLE, and RAUMZEIT programs are based on finite difference techniques whereas QX1 applies flux factorization.

The reactor calculated in this benchmark problem is a thermal reactor with the neutron velocity increased to correspond to that of a fast reactor.

The value of the flux amplitude, $p(0.005)$, is plotted as a function of the average length of the interval between shape calculations.

Superprompt-Critical Asymmetric Transient



Flux amplitude at 5 ms from a superprompt-critical asymmetric transient

- It is seen that the accuracy of the RAUMZEIT results rapidly deteriorates if the step length is increased over $\sim 5 \times 10^{-5}$ s.
- In QX1, a much smaller number of shape functions is required.
- The point at 10^{-3} s corresponds to five shape function recalculations in the 5-ms time interval.
- If actual fast reactor cross-sections were used, the number of shape function calculations at 10^{-3} s could have been < 5 because of the smaller overall flux shape distortion in a fast reactor.
- It should be emphasized that the error of the QX1 results is negative and asymptotically approaches the error of the point kinetics results.
- Reducing the number of shape recalculations to zero yields the point kinetics results.

- Thus, the quasi-static results are between the exact and the PRM result.
- If the point kinetics results were plotted onto this figure, a single point would be plotted at the full length of the interval.
- The results of the factorized approach would end up at this point if the interval between shape calculations were to be successively increased.

Superprompt-Critical Asymmetric Transient

The high degree of computational stability, i.e., movement from "exact results" to point reactor model results as the interval between shape calculations is increased, is unique to the flux factorization approach.

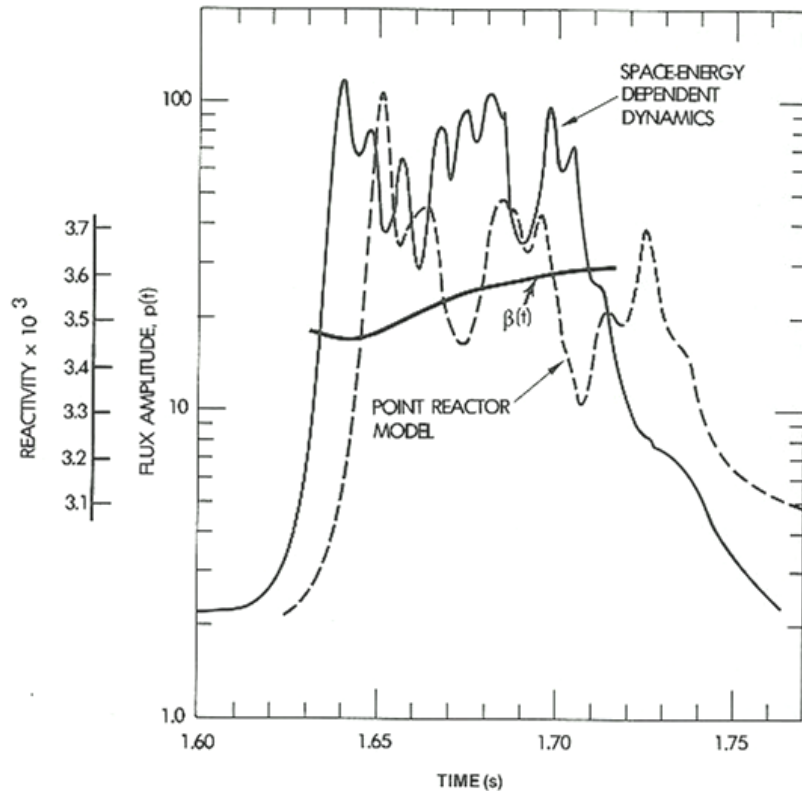
The error of the point kinetics treatment of reactor transients without feedback may be dramatic, possibly several orders of magnitude.

If, however, nonlinear feedback is included in a reactor transient, the error of the point kinetics treatment is generally substantially reduced.

For example, from several orders of magnitude without feedback to something like a factor of 2 or even less with feedback.

Amplitude during a Rapid Flow Coast-Down Accident Simulation in a Large Fast Reactor

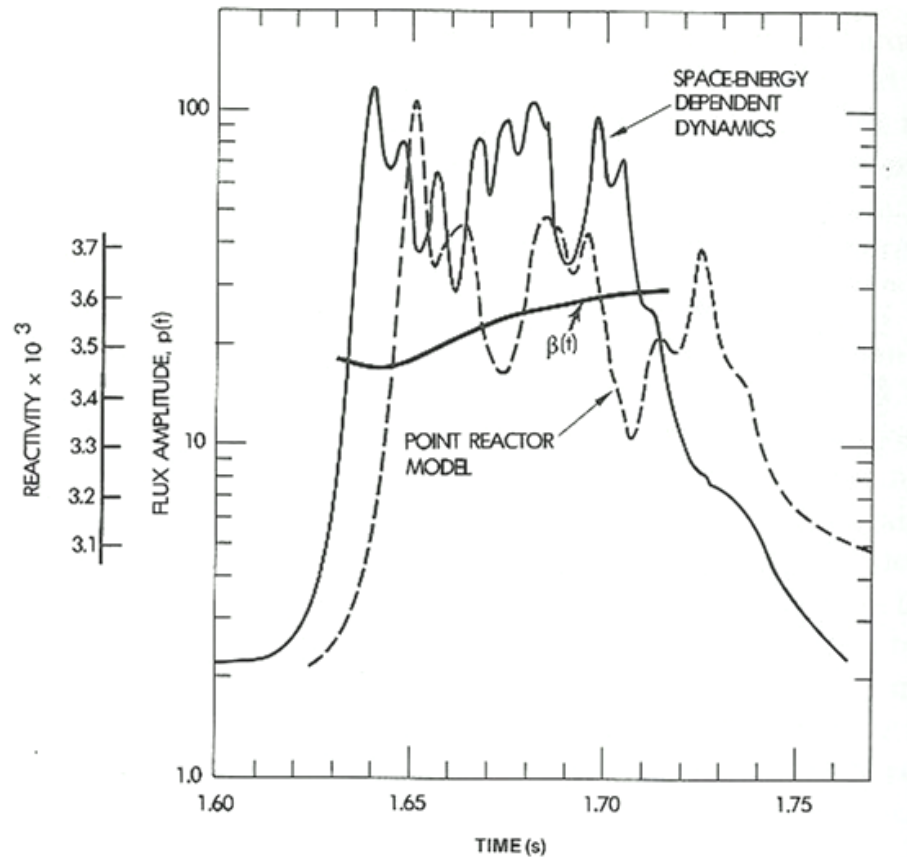
$$\rho_{\max} = 0.93\%$$



- The complicated structure of this transient is caused by the superposition of the reactivity that results from coolant boiling in various cooling channels.
- The solid curve is calculated with the QX2S space-energy dependent dynamics (SED) program.
- The PRM result is calculated with the same program by eliminating the shape recalculation.
- The SED flux in the pre-boiling phase is somewhat larger than the PRM flux.
- Therefore, boiling of the hottest channel starts somewhat earlier in the SED treatment, and the positive reactivity associated with sodium boiling drives the power up at an earlier time than in the PRM treatment.
- At 1.64 s, the SED flux is ~ 25 times larger than the PRM flux.
- However, negative Doppler feedback starts turning the flux down at about the same maximum value, i.e., after a comparable temperature rise as indicated by comparable integrals over the first burst.

Amplitude during a Rapid Flow Coast-Down Accident Simulation in a Large Fast Reactor

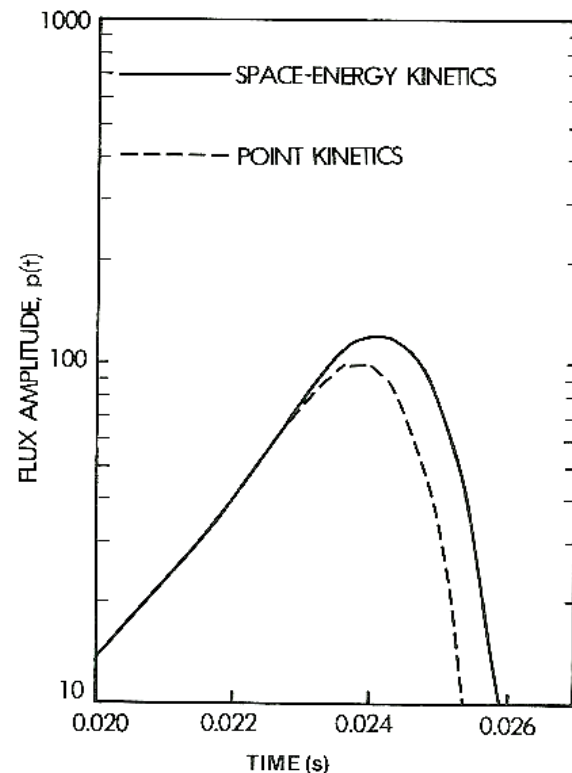
$$\rho_{\max} = 0.93\%$$



- The second burst is more energetic in the SED than in the PRM.
- As a consequence of this, boiling in the next channel starts earlier in SED and the entire further superposition of reactivity insertions is different.
- The PRM treatment stretches this transient out farther in time.
- The overall energy release in this example is $\sim 38\%$ larger utilizing SED than in the PRM model approximation.

Amplitude during a Core-Disruptive Accident Simulation in a Large Fast Reactor

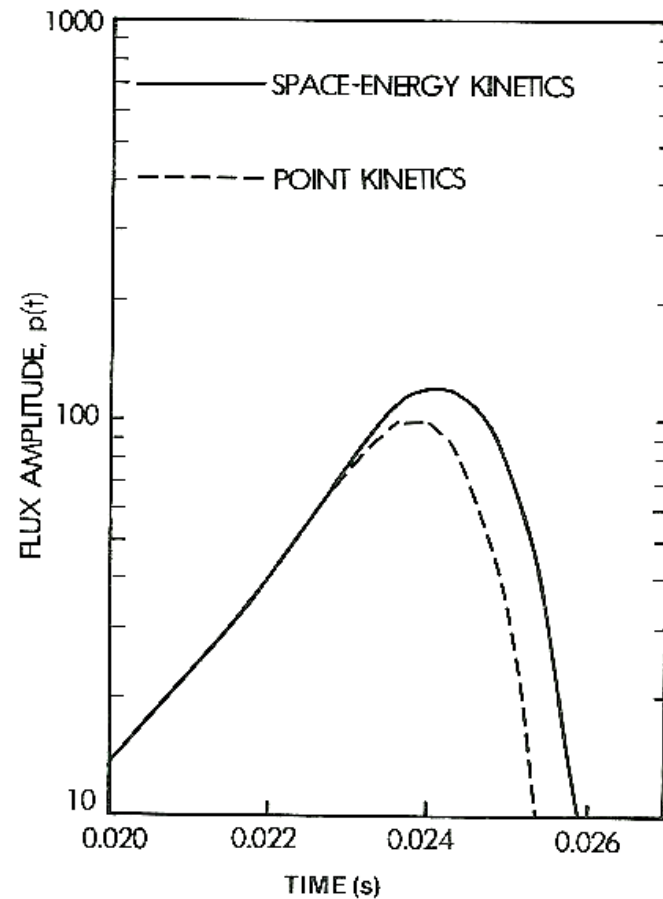
$$\rho_{\max} = 0.93\%$$



- Since core disassembly transients are shut down by displacement of fuel, significant differences between the SED and PRM results can be expected.
 - To isolate these differences during the actual disassembly, the "initial" temperature was assumed to be at the onset point of disassembly, and the reactivity was inserted by increasing ν .
 - This eliminates the differences between PRM and SED results during the rising phase of the transient.
 - Thus, the only differences that occur in this artificial example show up during the decreasing phase.
 - The SED results are higher than the PRM results due to a higher reactivity resulting from flux deformation.
-
- The moving fuel takes part of the flux with it, which results in a higher flux weight factor for the moved fuel than in the PRM.
 - The SED flux during this phase is a factor of 2 or more larger than the PRM result.

Amplitude during a Core-Disruptive Accident Simulation in a Large Fast Reactor

$$\rho_{\max} = 0.93\%$$



- The major part of the energy is released during the rising phase of the transient and around the maximum where the deviation between the two treatments is small in this special example.
- The transient decreases much faster than it increases due to the rapid disassembly acting in addition to the Doppler feedback effect (Doppler feedback alone leads to a symmetrical transient).
- A direct consequence of the asymmetry of the transient is that the energy production during the decreasing phase is smaller than during the power increase.
- Therefore, the inaccuracy of the treatment during the disassembly does not have a very large effect on the energy release since this inaccuracy affects only the smaller part of the overall energy production.

Dynamic Reactivity Coefficients

DYNAMIC REACTIVITY COEFFICIENTS

The reactivity coefficients are the classical point kinetics concepts that relate changes in a reactor to the corresponding variations of the reactivity.

With the SED dynamics, the automatic extensions of these concepts are "dynamic reactivity coefficients" .

SED dynamics using the factorizing approach involves the dynamic reactivity, $\rho(t)$, directly.

Other methods that aim at the calculation of the time-dependent flux can calculate the dynamic reactivity *a posteriori* by inserting the time-dependent macroscopic group cross-sections in the definition of $\rho(t)$ by the scalar product.

DYNAMIC REACTIVITY COEFFICIENTS

Various feedback phenomena are normally affecting $\rho(t)$ $\rightarrow$ reactivity coefficients are important for an evaluation of the integral feedback effects.

Applications of such reactivity coefficients are for:

- *reactor operation;*
- *pre-operational demonstration of safety characteristics;*
- *a posteriori separation of reactivity effects to better understand the transients involved in safety analysis, as the basis for design decisions on safety parameters;*
- *designing and analyzing experiments for an evaluation of the accuracy of computational predictions of the involved temperature and reactivity changes.*

DYNAMIC REACTIVITY COEFFICIENTS

In general, reactivity coefficients are defined by partial derivatives: $\gamma_{\xi} = \frac{\partial \rho}{\partial \xi}$

with ξ being the changing quantity.

If ξ is an integral quantity, as is ρ , the result of the partial derivative is an integral quantity for the reactor.

It may depend on ξ itself and on other parameters as well.

The typical example is the power coefficient, γ_p : $\gamma_p = \frac{\partial \rho}{\partial P} \approx \frac{\Delta \rho}{\Delta P}$

Note that:

The power coefficient is the major steady-state reactivity coefficient in which the various phenomena such as temperature effects, expansions throughout the system, xenon and samarium poisoning appear in steady-state combinations.

The power coefficient is not unique, as interim power levels may involve different coolant flow and thus different temperature conditions.

EXAMPLE: THE TEMPERATURE COEFFICIENT

The integral temperature coefficient, with a suitably defined temperature T, is given by:

$$\gamma_T = \frac{\partial \rho}{\partial T} \approx \frac{\Delta \rho}{\Delta T}$$

→ the resulting γ_T is not unique.

The temperature coefficient depends on the combination of the temperature field changes in the various reactor constituents, since fuel, coolant, and moderator reactivity contributions normally appear in different combinations, depending on the type of transient.

DYNAMIC REACTIVITY COEFFICIENTS

The combination of local contributions to reactivity changes and the space-dependent temperature field can be expressed in the following way:

If the energy integration of the scalar product is carried out first, one obtains the "local reactivity contribution (per volume element)," $\delta\rho(r)$, appearing as an integrand of the remaining spatial integration:

$$\Delta\rho = \int_V \delta\rho_V(r) dV$$

Expanding the integrand with the local temperature change, $\delta T(r)$ gives:

$$\Delta\rho = \int_V \frac{\delta\rho_V(r)}{\delta T_V(r)} \delta T(r) dV = \int_V \gamma_{TV} \delta T(r) dV$$

where the first factor in the integrand is the local temperature coefficient (again per volume element).

The integral quantity $\Delta\rho$ primarily affects the transient of the amplitude function.

The local contributions, $\Delta\rho_V(r)$, may distort the space distribution of the flux, the shape function.

DYNAMIC REACTIVITY COEFFICIENTS

Some Remarks:

1. PRM may significantly underpredict power transients and the resulting energy release.
2. On the other hand, the full SED treatment, even using the improved quasi-static approach, may be too lengthy and time consuming for routine application in some cases.
3. This suggests the desirability of further developments in the spatial dynamics area.
4. These developments consist of refinements of the quasi-static computation scheme, of augmentation of point reactor dynamics, or of "intermediate" approaches (between the PRM and complete SED dynamics).

DYNAMIC REACTIVITY COEFFICIENTS

More Remarks:

To speed up the improved quasi-static method, as applied to three-dimensional problems, we can:

1. Elaborate extrapolation schemes for the flux amplitude and the reactivity in order to stretch the reactivity and the shape function time steps.
2. Error-estimating procedures are employed to avoid unnecessary flux shape and reactivity calculations.
3. The time-dependent β_k calculations by inner products are replaced by simple empirical or theoretical formulas or at least performed much less frequently than the reactivity calculation, e.g., only at shape steps.
4. Reactivity coefficients with appropriate coupling can also be employed to stretch the intervals between the inner product and the shape calculations in the factorized spatial dynamics.
5. Then, both routes for developing intermediate approaches - speeding up the SED dynamics and augmenting the PRM - might eventually merge into a single method for the treatment of dynamics problems with any desired degree of accuracy.

Next Class: *High-Fidelity Neutronics, Thermal-Hydraulics
and Fuel Performance Models; Feedback
Parameters*